北京交通大學

信号与系统

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and Information Engineering

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Frequency domain analysis of signals

- Frequency domain analysis of continuous periodic signals
- Frequency domain analysis of continuous aperiodic signals
- Frequency domain analysis of discrete periodic signals
- Frequency domain analysis of discrete aperiodic signals
- Time domain sampling of continuous-time signals
- Analysis of signal spectrum by MATLAB

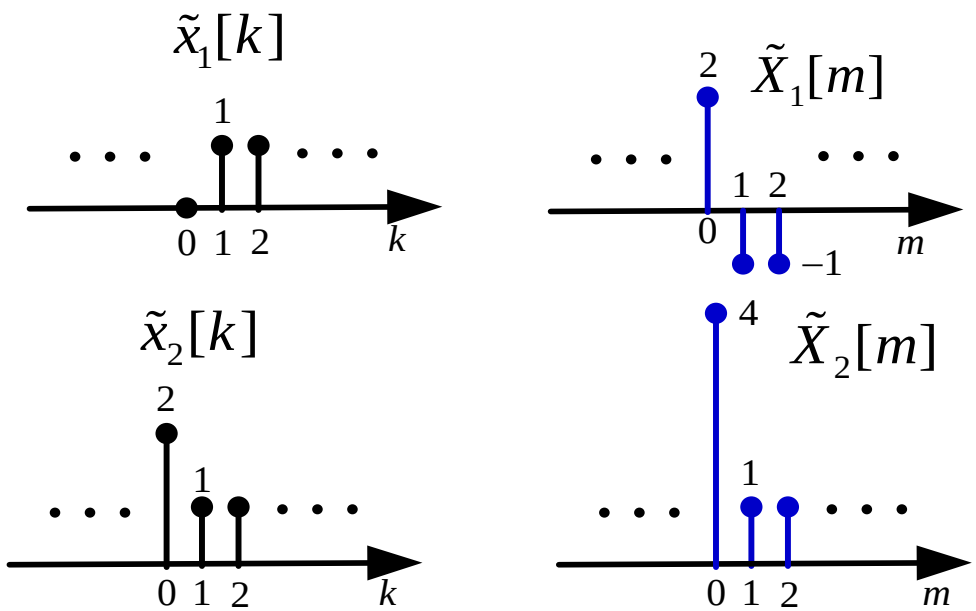
Frequency domain analysis of discrete periodic signals

- ※ Frequency domain representation of discrete periodic signals
- ※ Spectrum of discrete periodic signals
- ※ Properties of discrete periodic signal DFS

Frequency domain representation of discrete periodic signals

Discrete periodic signal with period N $\tilde{x}[k]$ can employ the imaginary exponential sequence to express as:

$$\tilde{x}[k] = \frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}[m]\mathrm{e}^{\mathrm{j}\Omega_m k} \qquad \Omega_m = \frac{2\pi}{N}m \qquad m = 0,1,\cdots,N-1$$



Different time domain signals have different weighting coefficients $\tilde{X}[m]$ in front of the imaginary exponential order.

Frequency domain representation of discrete periodic signals

IDFS $$\tilde{x}[k]=\frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}[m]\mathrm{e}^{\mathrm{j}\Omega_m k} \qquad \Omega_m=\frac{2\pi}{N}m \qquad m=0,1,\cdots,N-1$$

DFS $$\tilde{X}[m]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\Omega_m k}$$

$\tilde{X}[m]$ is called the spectrum of discrete periodic signals $\tilde{x}[k]$.

$$\tilde{x}[k]\xleftrightarrow{\text{DFS}}\tilde{X}[m]$$

Spectrum of discrete periodic signals

Features of $\tilde{X}[m]$:

(1) $\tilde{X}[m]$ is a discrete spectrum.

(2) $\tilde{X}[m]$ is a periodic sequence with a period of N.

$$\tilde{X}[m+N]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}(m+N)k}=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}=\tilde{X}[m]$$

$$\tilde{X}[m]=\underline{\left|\tilde{X}[m]\right|}\,\mathrm{e}^{\mathrm{j}\underline{\varphi[m]}}$$

Amplitude spectrum

Phase spectrum

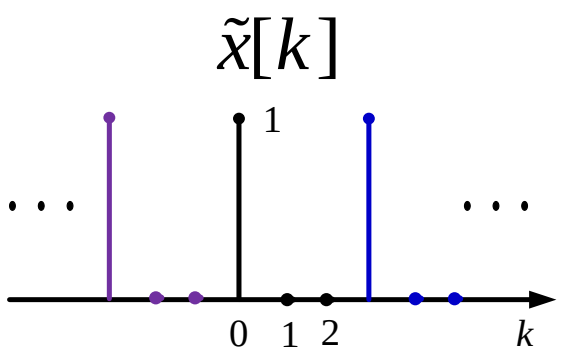
Spectrum of discrete periodic signals

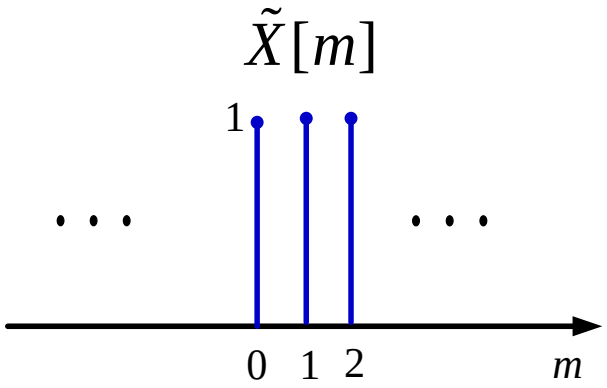
Periodic unit pulse sequence

Find the frequency spectrum of the periodic unit pulse sequence with a period of 3 as shown in the figure.

Solution:

$$\tilde{X}[m]=\sum_{k=0}^{2}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}mk}=1\cdot\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}m\cdot 0}=1$$





Spectrum of discrete periodic signals

Ex.: Find the frequency spectrum of a sequence $\tilde{x}[k]=\{\cdots,\overset{\downarrow}{0},1,1,\cdots\}$ with period 3.

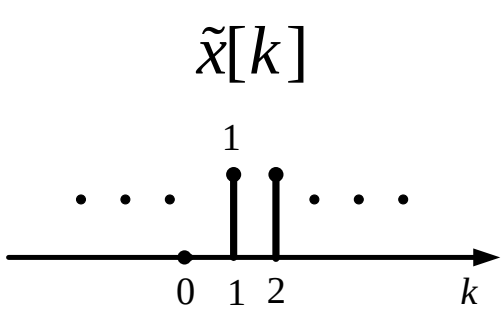
Solution:

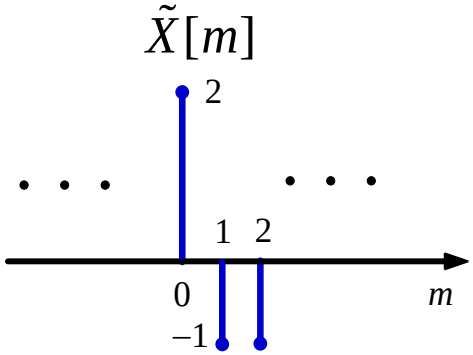
$$\tilde{X}[m]=\sum_{k=0}^{2}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}mk}$$

$$\tilde{X}[0]=\tilde{x}[0]+\tilde{x}[1]+\tilde{x}[2]=2$$

$$\tilde{X}[1]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}\cdot 1\cdot 1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}\cdot 1\cdot 2}=-1$$

$$\tilde{X}[2]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}\cdot 2\cdot 1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}\cdot 2\cdot 2}=-1$$





Spectrum of discrete periodic signals

Ex.: Find the frequency spectrum of a sequence $\tilde{x}[k]=\{\cdots,\overset{\downarrow}{1},2,3,4,\cdots\}$ with a period of 4.

Solution:

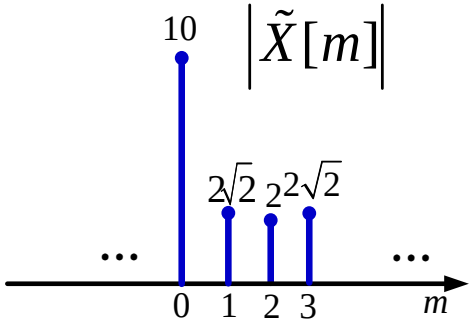
$$\tilde{X}[m]=\sum_{k=0}^{3}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}mk}$$

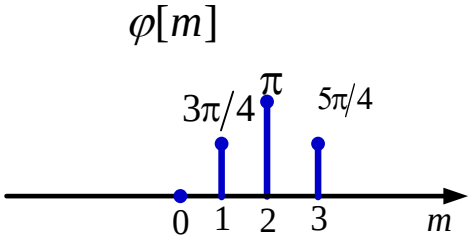
$$\tilde{X}[0]=\tilde{x}[0]+\tilde{x}[1]+\tilde{x}[2]+\tilde{x}[3]=10$$

$$\tilde{X}[1]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 1\cdot 1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 1\cdot 2}+\tilde{x}[3]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 1\cdot 3}=-2+2\mathrm{j}$$

$$\tilde{X}[2]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 2\cdot 1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 2\cdot 2}+\tilde{x}[3]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 2\cdot 3}=-2$$

$$\tilde{X}[3]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 3\cdot 1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 3\cdot 2}+\tilde{x}[3]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot 3\cdot 3}=-2-2\mathrm{j}$$





Spectrum of discrete periodic signals

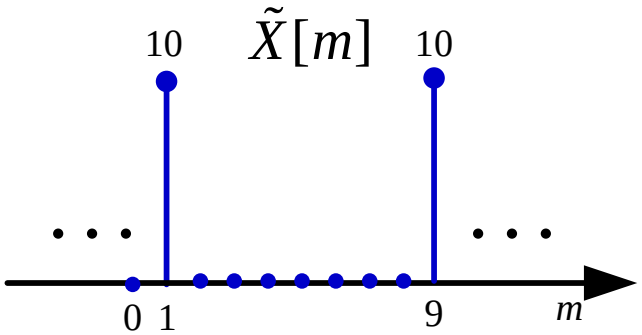
Ex.: Find the frequency spectrum of a periodic sequence $\tilde{x}[k]=2\cos\left[\pi k/5\right]$

Solution: The period of a periodic sequence $\tilde{x}[k]=2\cos\left[\pi k/5\right]$ is 10.

$$\tilde{x}[k]=\frac{1}{10}\left(10\mathrm{e}^{\mathrm{j}\frac{2\pi}{10}k}+10\mathrm{e}^{-\mathrm{j}\frac{2\pi}{10}k}\right)=\frac{1}{10}\left[10\mathrm{e}^{\mathrm{j}\frac{2\pi}{10}k}+10\mathrm{e}^{\mathrm{j}\frac{2\pi}{10}(10-1)k}\right]$$

Compared with IDFS expression, the DFS coefficient of the periodic sequence $\tilde{x}[k]$ can be obtained as:

$$\tilde{X}[m]=\begin{cases}10 & m=1,\ 9\\ 0 & m=0,2\le m\le 8\end{cases}$$



Spectrum of discrete periodic signals

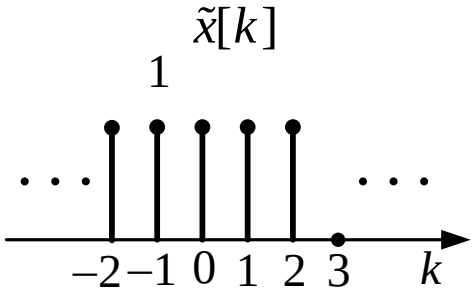
Example: Find the frequency spectrum of a periodic rectangular sequence with a period of 6 as shown in the figure.

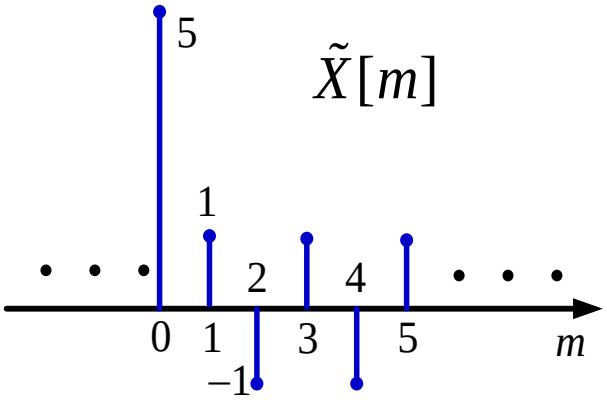
Solution:

$$\tilde{X}[m] = \sum_{k=-2}^{3} \tilde{x}[k] \mathrm{e}^{-\mathrm{j}\Omega_m k}$$

$$= \frac{\mathrm{e}^{\mathrm{j}\frac{2\pi}{6}2m} - \mathrm{e}^{-\mathrm{j}\frac{2\pi}{6}3m}}{1 - \mathrm{e}^{-\mathrm{j}\frac{2\pi}{6}m}}$$

$$= \frac{\sin\left(\frac{5\pi}{6}m\right)}{\sin\left(\frac{\pi}{6}m\right)}$$





Properties of discrete periodic sequence DFS

1. Linear characteristics

$$\mathrm{DFS}\{a\tilde{x}_1[k]+b\tilde{x}_2[k]\}=a\mathrm{DFS}\{\tilde{x}_1[k]\}+b\mathrm{DFS}\{\tilde{x}_2[k]\}$$

Properties of discrete periodic sequence DFS

Ex.: Find the frequency spectrum of a periodic sequence $\tilde{x}[k]=\{\cdots,\overset{\downarrow}{2},1,1,\cdots\}$ with a period of 3.

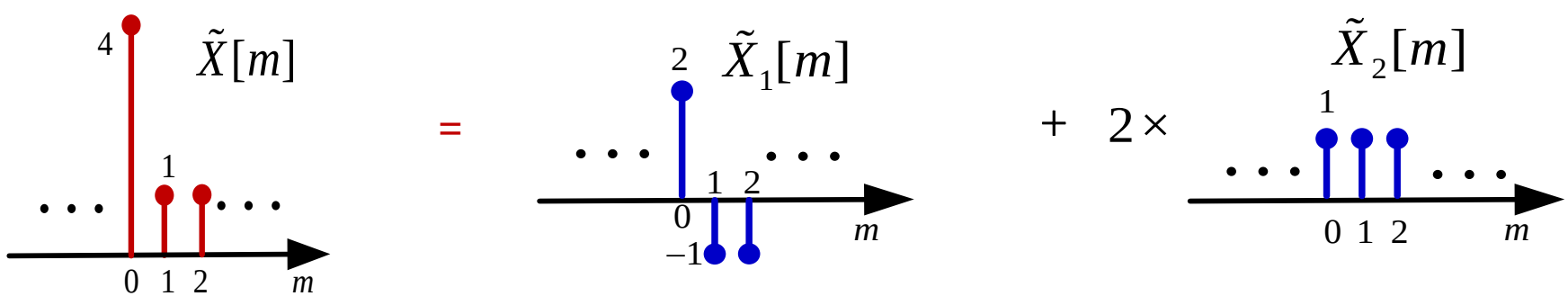
$$\tilde{x}_1[k]=\{\cdots,\overset{\downarrow}{0},1,1,\cdots\}\xleftrightarrow{\text{DFS}}X_1[m]=\{\cdots\overset{\downarrow}{2},-1,-1,\cdots\}$$

$$\tilde{x}_2[k]=\{\cdots,\overset{\downarrow}{1},0,0,\cdots\}\xleftrightarrow{\text{DFS}}X_2[m]=\{\cdots\overset{\downarrow}{1},1,1,\cdots\}$$

Solution: Represent $\tilde{x}[k]$ as $\tilde{x}_1[k]+2\cdot\tilde{x}_2[k]$

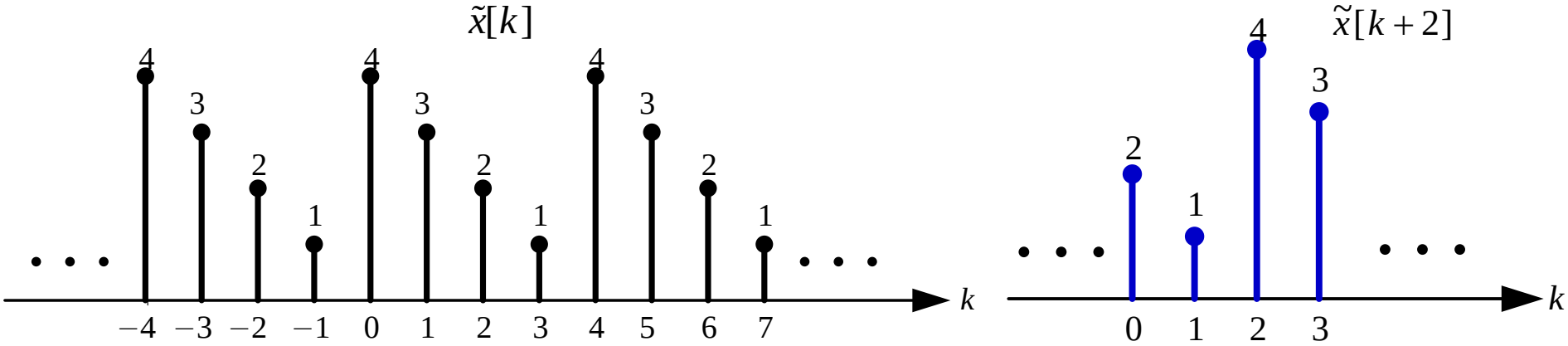
By the linear characteristics of DFS, it can be obtained

$$\tilde{X}[m]=\tilde{X}_1[m]+2\cdot\tilde{X}_2[m]$$



Properties of discrete periodic sequence DFS

2. Displacement characteristics



Displacement of periodic sequence

After displacement, the periodic sequence is still a periodic sequence with the same period, so it is only necessary to observe the situation of the sequence after displacement for one period.

Properties of discrete periodic sequence DFS

2. Displacement characteristics

(a) Displacement characteristics in time domain

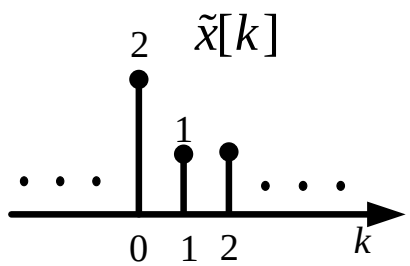
$$\mathrm{DFS}\left\{\tilde{x}[k-n]\right\} = \tilde{X}[m]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mn}$$

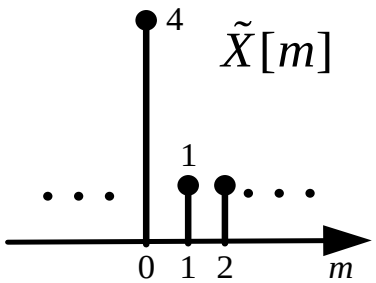
The displacement of a sequence in time domain corresponds to its phase shift in frequency domain.

Properties of discrete periodic sequence DFS

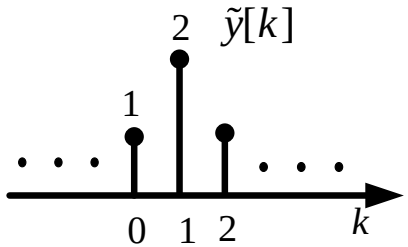
Example: Given a sequence $\tilde{x}[k]=\{\cdots,\overset{\downarrow}{2},1,1,\cdots\}$ with a period of 3, find the spectrum of $\tilde{y}[k]=\tilde{x}[k-1]$.

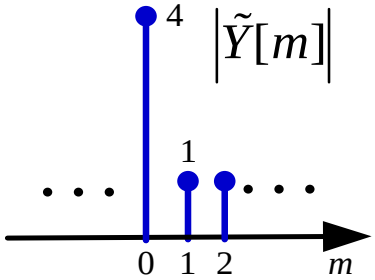
Solution:

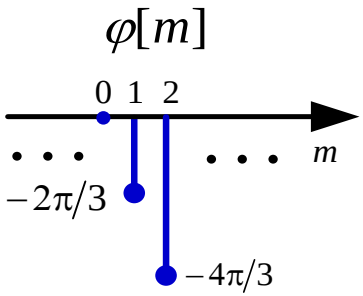




$$\tilde{Y}[m]=\tilde{X}[m]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{3}\cdot m}$$







Properties of discrete periodic sequence DFS

2. Displacement characteristics

(b) Displacement characteristics in frequency domain

$$\mathrm{DFS}\left\{\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}lk}\right\} = \tilde{X}[m+l]$$

The phase shift of a sequence in time domain corresponds to its displacement in frequency domain.

Properties of discrete periodic sequence DFS

Ex.: Given a sequence $\tilde{x}[k]=\{\cdots,\overset{\downarrow}{2},1,1,\cdots\}$ with a period of 3, find the spectrum of $\tilde{y}[k]=\tilde{x}[k]\cos(2\pi k/3)$.

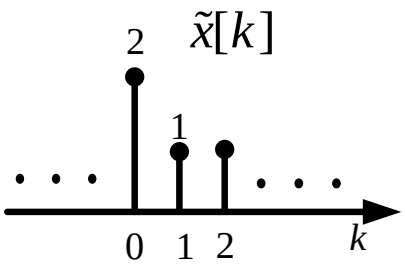
Solution:

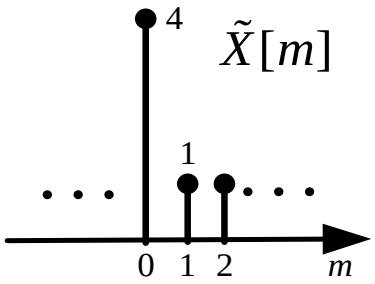
$$\tilde{y}[k]=\frac{\tilde{x}[k]}{2}\cdot\left(\mathrm{e}^{\mathrm{j}2\pi k/3}+\mathrm{e}^{-\mathrm{j}2\pi k/3}\right)$$

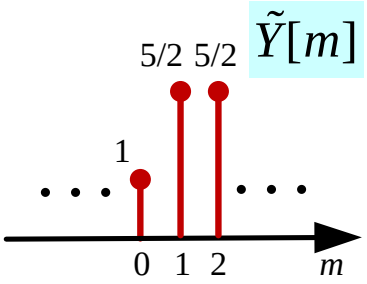
$$\tilde{Y}[m]=\frac{1}{2}\left\{\tilde{X}[m-1]+\tilde{X}[m+1]\right\}$$

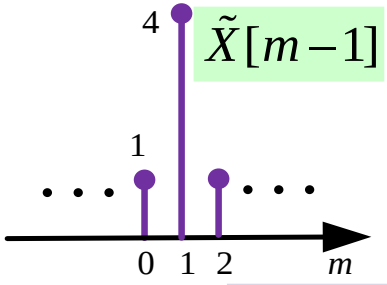
Properties of discrete periodic sequence DFS

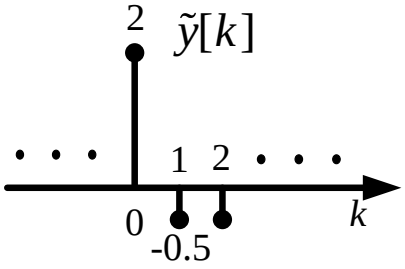
$$\tilde{Y}[m]=\frac{1}{2}\left\{\tilde{X}[m-1]+\tilde{X}[m+1]\right\}$$

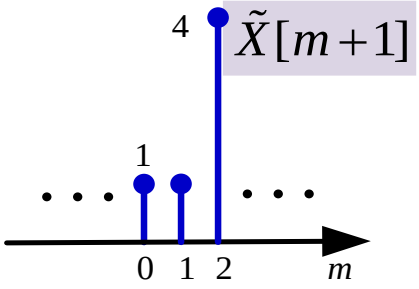










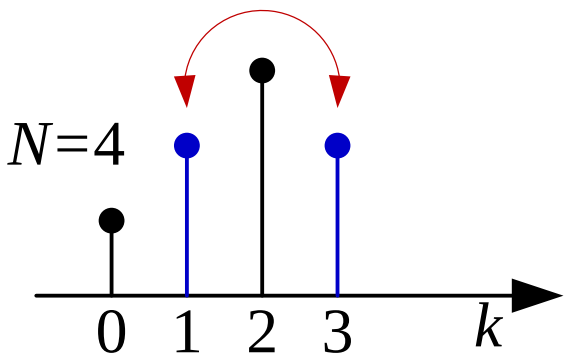


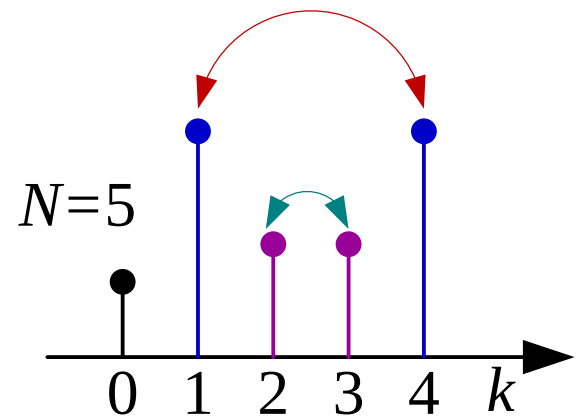
Properties of discrete periodic sequence DFS

3. Symmetry characteristics

Even symmetry of periodic sequences

$$\tilde{x}[k] = \tilde{x}[-k] = \tilde{x}[N-k]$$



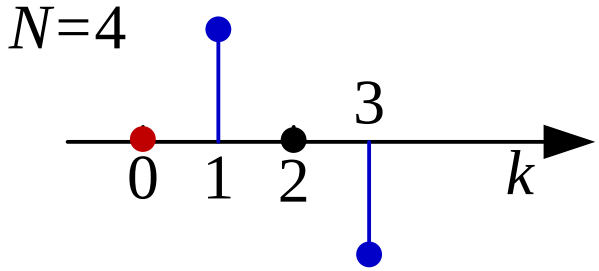


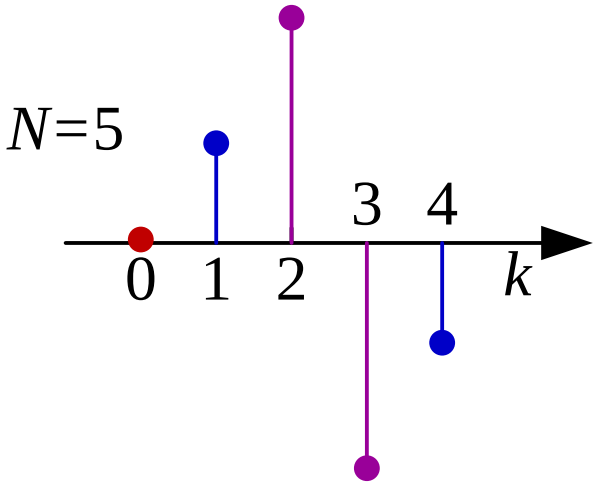
Properties of discrete periodic sequence DFS

3. Symmetry characteristics

Odd symmetry of periodic sequences

$$\tilde{x}[k] = -\tilde{x}[-k] = -\tilde{x}[N-k]$$





Properties of discrete periodic sequence DFS

3. Symmetry characteristics

$$\mathrm{DFS}\{\tilde{x}^*[k]\} = \tilde{X}^*[-m]$$

$$\mathrm{DFS}\{\tilde{x}^*[-k]\} = \tilde{X}^*[m]$$

If $\tilde{x}[k]$ is a **real sequence**, there are

$$\tilde{X}[m] = \tilde{X}^*[-m]$$

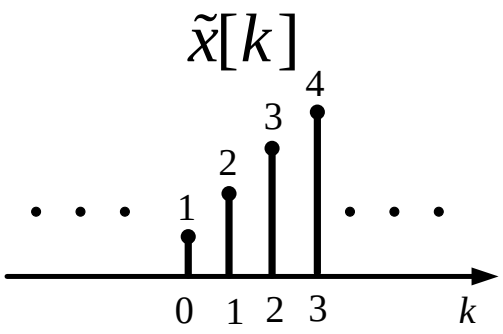
$$\left|\tilde{X}[m]\right| = \left|\tilde{X}[-m]\right|$$

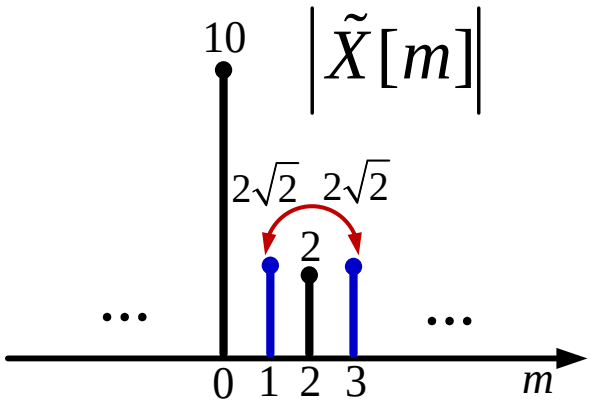
$$\varphi[m] = -\varphi[-m]$$

$$\tilde{X}_{\mathrm{R}}[m] = \tilde{X}_{\mathrm{R}}[-m]$$

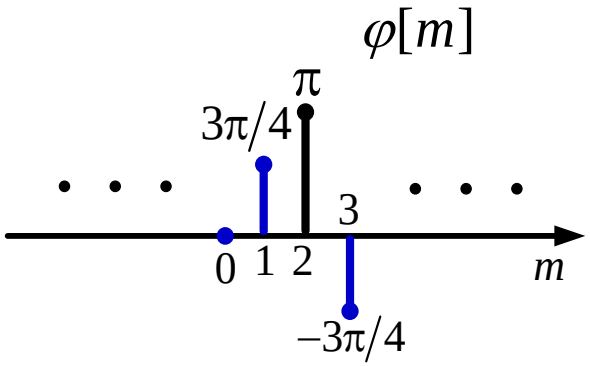
$$\tilde{X}_{\mathrm{I}}[m] = -\tilde{X}_{\mathrm{I}}[-m]$$

Properties of discrete periodic sequence DFS





If $\tilde{x}[k]$ is a real sequence, the amplitude spectrum is even symmetric and the phase spectrum is odd symmetric.



Properties of discrete periodic sequence DFS

4. Periodic convolution characteristics

※ Time domain periodic convolution theorem:

$$\mathrm{DFS}\left\{\tilde{x}_1[k] \tilde{*} \tilde{x}_2[k]\right\} = \mathrm{DFS}\{\tilde{x}_1[k]\} \bullet \mathrm{DFS}\{\tilde{x}_2[k]\}$$

※ Frequency domain periodic convolution theorem:

$$\mathrm{DFS}\left\{\tilde{x}_1[k] \bullet \tilde{x}_2[k]\right\} = \frac{1}{N} \mathrm{DFS}\{\tilde{x}_1[k]\} \tilde{*} \mathrm{DFS}\{\tilde{x}_2[k]\}$$

The periodic convolution of time domain corresponds to the product of frequency domain;

The product of time domain corresponds to the periodic convolution of frequency domain.

Properties of discrete periodic sequence DFS

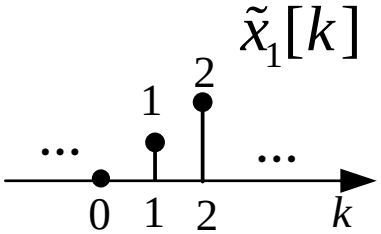
Definition of periodic convolution:

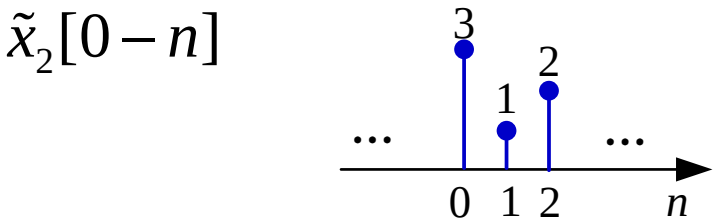
$$\tilde{x}_1[k] \tilde{*} \tilde{x}_2[k] = \sum_{n=0}^{N-1} \tilde{x}_1[n] \tilde{x}_2[k-n]$$

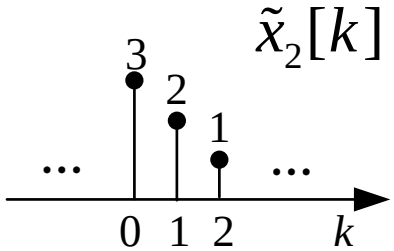
※ Periodic convolution is the convolution operation of two periodic sequences with equal periods.

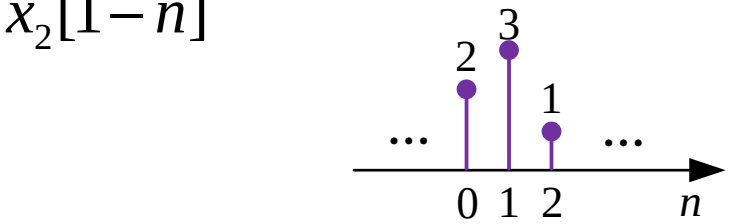
※ The result of periodic convolution is still a periodic sequence with the same period.

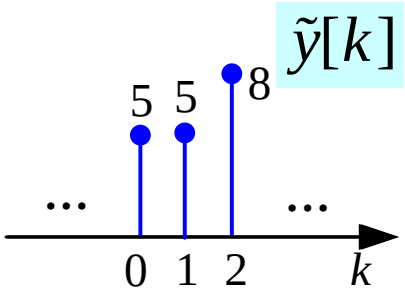
Example: The sequences $\tilde{x}_1[k], \tilde{x}_2[k]$ with a period of 3 is shown in the figure. Calculate $\tilde{y}[k] = \tilde{x}_1[k] \tilde{*} \tilde{x}_2[k]$.

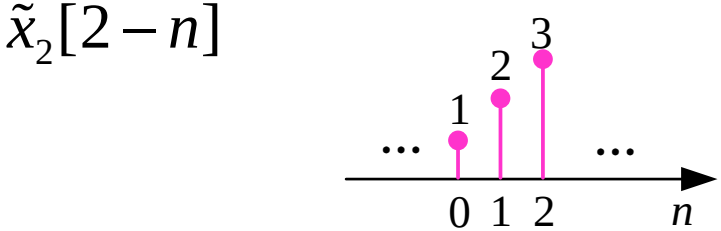












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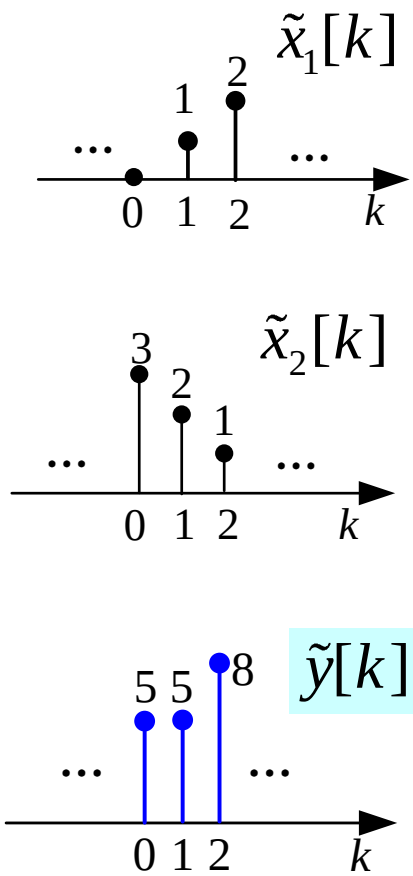

Matrix representation of periodic convolution：

$$\tilde{x}_1[k] \tilde{*} \tilde{x}_2[k] = \sum_{n=0}^{N-1} \tilde{x}_1[n] \tilde{x}_2[k-n]$$

$$\begin{bmatrix} \tilde{y}[0] \\ \tilde{y}[1] \\ \tilde{y}[2] \end{bmatrix} = \begin{bmatrix} \tilde{x}_2[0] & \tilde{x}_2[-1] & \tilde{x}_2[-2] \\ \tilde{x}_2[1] & \tilde{x}_2[0] & \tilde{x}_2[-1] \\ \tilde{x}_2[2] & \tilde{x}_2[1] & \tilde{x}_2[0] \end{bmatrix} \begin{bmatrix} \tilde{x}_1[0] \\ \tilde{x}_1[1] \\ \tilde{x}_1[2] \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 8 \end{bmatrix}$$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!